

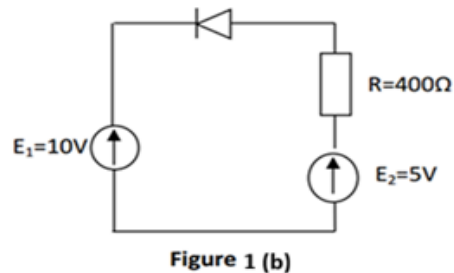
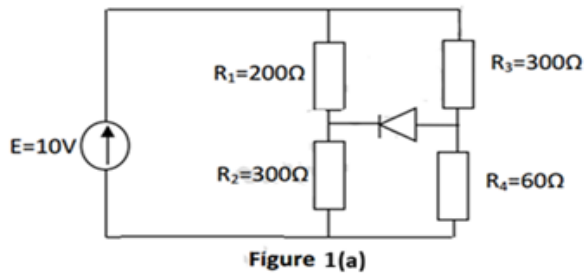
Exercise Series : Diode and Zener Diode

Exercise 1 (course)

Determine the conducting or cutoff state (ON or OFF) of the diode for the following circuits (Fig1 a, b).

- In the event that the diode is conducting, find the current I flowing through it.

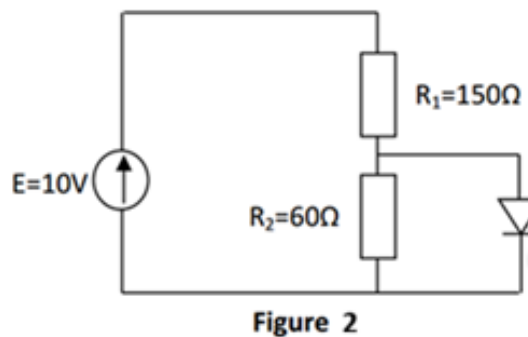
Assume that the diode is ideal with a threshold voltage of 0.7 V.



Exercise 2

In the following circuit Fig.2:

- 1- determine whether the diode is in the conducting (ON) or cutoff (OFF) state.
- 2- If the diode is conducting, determine the current I flowing through it. Assume the diode is ideal with a threshold voltage (knee voltage) of 0.7 V.



Exercise 3 (course)

In the circuit shown in Figure (4), the diode is assumed to be ideal ($V_0 = 0.7V$, $r_d = 0\Omega$).

Calculate V_{R_2} for the following cases:

- 1- $E = 5V$
- 2- $E = -5V$
- 3- $e(t) = 5 \sin(\frac{2\pi}{T}t)$ with $T=20$ ms

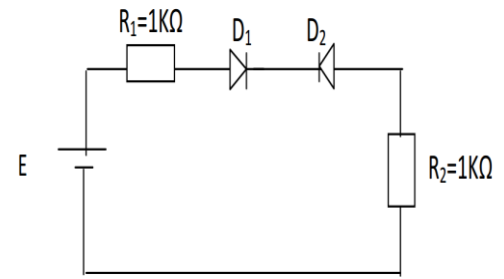


Figure.3

Exercise 4

Given the following circuit Fig4:

I- Replace the circuit seen between terminals A and B with its Thevenin equivalent generator, and calculate E_{th} and R_{th} .

II- Calculate the current I for the following three cases:

1. The diode is ideal ($V_D = 0V$).
2. The diode has a threshold voltage ($V_D = 0.7V$).
3. The diode has a threshold voltage and an internal resistance (R_D).

Given values: $E=12V$, $R_1=6$ kΩ, $R_2=3$ kΩ, $R_C=1$ kΩ et $R_D=100$ Ω.

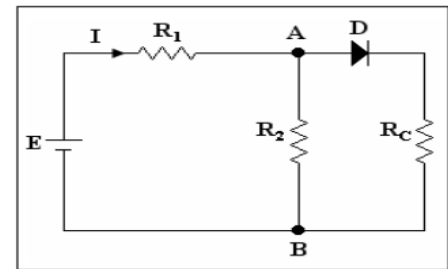


Figure 4

Exercise 5(course)

Consider the following circuit:

$e(t) = 5 \sin(\frac{2\pi}{T}t)$ with: $T=20$ ms, $V_{D1}=V_{D2}=0.6V$ and $R_1=R_2=1$ kΩ

- Determine the voltage across R_2 (U_{R2}).

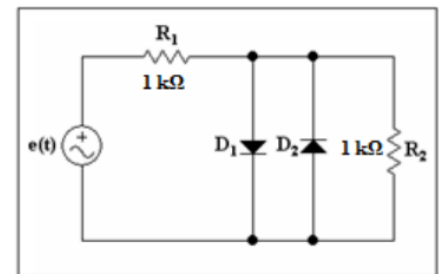


Figure 5

Exercise 6

Consider the following diagrams:

$V_D = 0.6V$ and $R_1 = R_2 = 1$ kΩ.

- 1-For $E = 5$:

Calculate the voltage across R_2 (U_{R2}).

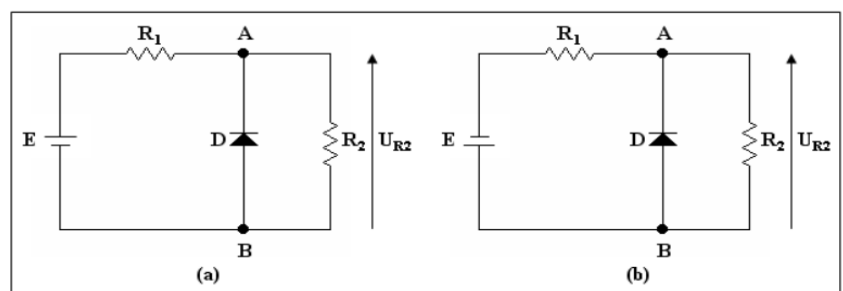
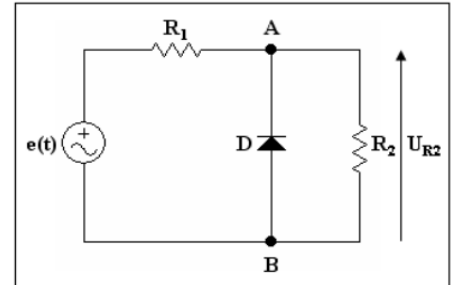


Figure 6

2-The DC voltage source E is replaced by an Alternating voltage generator (AC source):

$$e(t) = 5 \sin\left(\frac{2\pi}{T}t\right), \text{ avec : } T=20 \text{ ms.}$$

- Determine the voltage across R2 (U_{R2}).



Exercise 7 (course)

The datasheet of a Zener diode provides the following:

$V_Z = 12V$, $I_Z = 50mA$, $I_{Zmin} = 0,5mA$, $I_{Zmax} = 100mA$ and $R_Z = 20\Omega$

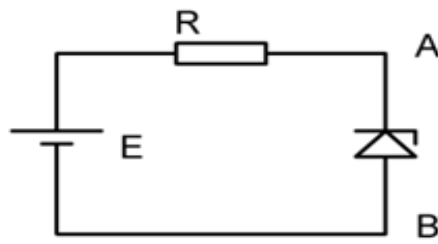
1. Calculate the voltages V_{Zmin} and V_{Zmax}
2. Determine the equation for the characteristic (the straight line) of the Zener diode.
3. Calculate the voltages V_{Z1} et V_{Z2} for the respective currents: $I_{Z1} = 20mA$ et $I_{Z2} = 80mA$

Exercise 8

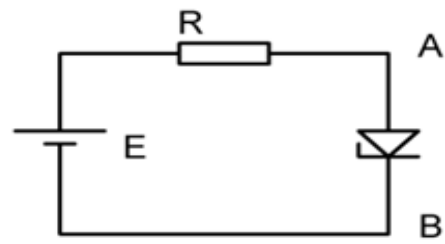
Consider the following figures:

$R = 1k\Omega$ $V_Z = 6,2V$ $V_D = 0,6V$

1. Determine the voltage (U_{AB}) between points A and B (Figure 8(a)) for:
 - a. $E = 12V$
 - b. $E = 5V$
2. Determine the voltage (U_{AB}) between points A and B (Figure 8(b)) for:
 - a. $E = 5V$
 - b. $E = -12V$



(a)



(b)

Exercise 9

Consider the following circuit. The Zener diode D_z is assumed to be ideal. Its Zener voltage is equal to 6.2 V, and its maximum power rating is 1.3 W.

Given: $R = 100\ \Omega$.

1. Calculate the maximum current that can flow through the diode.
2. Let $R_L = 100\ \Omega$. Determine the range of variation of the supply voltage V_s that allows obtaining a stabilized output voltage V_z .
3. For the case $V_s = 30\text{ V}$, determine the range of variation of R_L that allows obtaining a stabilized voltage V_z .

